ARC-SPEC-2026-EDU-02: Laboratory Verification & Automated Harness Architecture

A Standardized Benchmark Suite for Sub-12ms Bare-Metal Admissibility Testing

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Status: Archive-Ready / Path A Academic Extension
Governing Invariants: $C_{\text{ops}} = 0$ | $\tau_{\text{override}} \leq 11.99\text{ms}$ | Heap Allocation = 0 bytes
Target Baseline: Phase 7 Realization (`v1.3.0-exec`)

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1 Executive Summary & Testing Philosophy

If ARC-SPEC-2026-EDU-01 establishes the mathematical and theoretical curriculum for Deterministic Admissibility Engineering (DAE), this specification defines the experimental and verification apparatus.

In traditional computer science education, student labs are evaluated in high-jitter, managed environments (such as Docker containers or JVM test runners) that mask race conditions, non-deterministic latency spikes, and heap fragmentation. In contrast, DAE laboratory verification operates under an absolute constraint:

$$\text{VERIFICATION RULE: } C_{\text{eval}} = 0 \quad \wedge \quad \text{WCET} \leq 11.99\text{ms}$$

This paper specifies `ARC-HARNESS-2026`—a zero-heap, containerless, hardware-in-the-loop (HIL) verification suite designed to run asynchronously inside `substrate-ci.yml`. It provides four standardized, auto-graded laboratory benchmarks that test whether student code maintains state vector stability, zero-copy packet ingress, and non-collapsible lattice ordering under extreme stress.

Harness Architecture Comparison

TRADITIONAL GRADED TEST RUNNER (Permissive / High Drag):

[Student Submission] $\longrightarrow$ [JVM/Python Harness] $\longrightarrow$ [Unbounded Heap] $\longrightarrow$ [Rubric]

DAE AUTOMATED VERIFICATION HARNESS (ARC-HARNESS-2026):

$$\text{[Student Submission]} \longrightarrow \text{[\#![no_std] Bare-Metal Runner]} \longrightarrow \text{[Benchmark]}$$
$$\Downarrow$$
$$\longrightarrow \text{[WCET } > 11.99\text{ms]} \longrightarrow \text{FAIL (POSIX 40)}$$

2 The Four Standardized DAE Benchmark Harnesses

2.1 Benchmark A: XDP Ring-0 Ingress & Frame Validation (LAB-XDP-01)

- **Objective:** Verify that student-written eBPF/XDP ingress filters intercept, parse, and validate incoming 64-byte frame HMACs at Ring 0 without allocating heap memory or passing malformed frames into user-space buffers.

Table 1: DAE Laboratory Benchmark Matrix

Benchmark	Core Test Objective	Mathematical Invariant Boundary
LAB-XDP-01	Ring-0 Frame Validation	$M_{\text{alloc}} = 0$ bytes $\quad \wedge \quad t_{\text{validation}} \leq 0.45\text{ms}$
LAB-VEC-02	5D State Vector Drift	If $\dot{\mathcal{T}} > 50$ ticks $\implies \mathcal{L}_{\text{eval}}(\mathbf{\Psi}) = \text{FREEZE}$ (POSIX 10)
LAB-HIL-03	HIL Motor Interlock	$\tau_{\text{override}} \leq 11.99\text{ms} \quad \wedge \quad \text{Torque}_{\text{actuator}} \to 0$ upon FAIL
LAB-QDX-04	Kingman Queue Clamping	Queue Length $Q_{\text{len}} \leq N_{\text{max_ring}}$ via PWC (POSIX 12)

- **Injected Stress Conditions:** Ingress rate at 10,000,000 packets/sec (line-rate saturation); 30% corrupted HMAC signatures; 20% oversized varint headers.

```rust
#[no_mangle]
pub fn verify_xdp_ingress(ctx: &XdpContext) -> XdpResult {
    let frame = parse_fixed_64byte_frame(ctx)?;

    // Invariant Check 1: Zero Heap Rule
    if HEAP_ALLOCATOR.has_allocated() {
        return XdpResult::Drop(POSIX_REASON_MEMORY_VIOLATION); // POSIX
            30
    }

    // Invariant Check 2: HMAC Validation Latency
    let start_cycles = read_tsc();
    let valid = verify_hmac_tcm(&frame);
    let elapsed_ms = cycles_to_ms(read_tsc() - start_cycles);

    if elapsed_ms > 0.45 {
        return XdpResult::Drop(POSIX_REASON_LATENCY_EXCEEDED); // POSIX
            32
    }

    if valid { XdpResult::Pass } else { XdpResult::Drop(
       POSIX_REASON_AUTH_FAILURE) }
}
```

Listing 1: LAB-XDP-01 Verification Logic (Rust Pseudocode)

2.2 Benchmark B: 5D Dynamic Vector Drift Tracking (LAB-VEC-02)

- **Objective:** Test the student's implementation of the 5D Dynamic State Vector $\mathbf{\Psi}_{\text{Entity}}(t)$ under simulated hardware clock drift ($\mathcal{T}$) and agency expansion ($\mathcal{A}$).
- **Evaluation Standard:** Student code must detect state drift velocity $V_a(t) = \frac{d\mathbf{\Psi}}{dt}$ and emit a POSIX 10 FREEZE mask before the state vector escapes its bounded constraint envelope when tick skew exceeds 50 ticks.

2.3 Benchmark C: Hardware-in-the-Loop Motor Interlock (LAB-HIL-03)

- **Objective:** Evaluate real-time physical safety interlocks by bridging software state evaluation with mechatronic hardware simulation.
- **Timeline Target:** From illegal toolpath command issuance (0.00ms) to lattice detection (2.50ms) and hardware NMI trigger (6.10ms), physical actuator torque must drop to 0 by ≤ 11.99ms.

2.4 Benchmark D: Kingman Queue Clamping & PWC (LAB-QDX-04)

- **Objective:** Test queue stability under non-linear packet arrival bursts ($\rho \geq 0.95$). Student code must engage Pending Worker Clearance (PWC / POSIX 12) throttling at $\rho = 0.80$ and execute wire-speed `XDP_DROP` filters at $\rho \geq 0.95$.

3 The Automated Execution Pipeline (`substrate-ci.yml`)

To maintain $C_{\text{ops}} = 0$, student submissions are processed asynchronously through a single GitHub Actions workflow.

```
# 1. Setup Rust Bare-Metal Target
rustup target add thumbv7em-none-eabi x86_64-unknown-none

# 2. Run Zero-Heap Allocation Audit
cargo build --release --target x86_64-unknown-none
valgrind --tool=dhat target/release/lab_submission

# 3. Execute Worst-Case Execution Time (WCET) Benchmark
cargo test --test lab_hil_benchmark -- --nocapture
python3 /scripts/validate_wcet_ceiling.py --max-limit-ms 11.99
```

Listing 2: `substrate-ci.yml` Execution Commands

4 Cryptographic Badging & POSIX Status Codes

When a submission completes all four benchmarks inside `substrate-ci.yml`, the runner generates a signed JSON web certificate anchored to the student's GitHub commit hash.

Table 2: System Status Codes & Error Mappings

Code	Status Label	Primary Root Cause
POSIX 0	PASS / ADMISSIBLE	All invariant benchmarks satisfied
POSIX 10	FREEZE / HOLD	Temporal clock skew exceeded (> 50 ticks)
POSIX 12	PWC / CLEARANCE	Pending Worker Clearance; token hold state
POSIX 30	MEMORY_ALLOCATION_VIOLATION	Heap memory allocated in `#![no_std]` context
POSIX 32	LATENCY_CEILING_EXCEEDED	Worst-Case Execution Time exceeded 11.99ms
POSIX 40	HARDWARE_SAFETY_FAIL	Dominance lattice violation / NMI override

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SPECIFICATION COMPLETE: ARC-SPEC-2026-EDU-02
DEPLOYMENT POSTURE: ARCHIVED FOR FUTURE PATH A EXTENSION
AUTHORIZATION HASH: A-77-DELTA-SHIELD-LOCKED